

Sports Betting Arbitrage Detection and Analysis System

Dominic Esposito
DSCI 510

May 2026

1 Motivation and Rationale

The goal of this project was to build a complete data pipeline capable of identifying sports betting arbitrage opportunities across multiple sportsbooks in near real time. Arbitrage betting occurs when odds discrepancies between sportsbooks allow a bettor to place wagers on all possible outcomes of an event and guarantee a mathematical profit regardless of the result.

This project was interesting because sportsbooks continuously update odds independently, creating temporary inefficiencies in the betting market. Detecting these opportunities requires collecting large amounts of live sports data, normalizing inconsistent formats between sportsbooks, integrating multiple datasets, and performing automated mathematical analysis at scale.

The project also provided an opportunity to apply concepts learned throughout the course, including web scraping, API usage, data cleaning, data integration, visualization, and statistical analysis. Additionally, this project demonstrated how automated data systems can continuously monitor changing information sources and generate actionable insights.

2 Description of Data Sources

The project collected betting odds data from several sportsbooks using a combination of APIs and Selenium-based web scraping.

2.1 Data Sources

The following sportsbooks were used:

- DraftKings
- BetMGM
- FanDuel
- Bovada
- BetRivers

The project monitored odds for the following sports:

- NBA
- MLB
- NHL

2.2 Data Collection Methods

Two different collection methods were used:

1. **The Odds API** was used to retrieve structured sportsbook odds data for FanDuel, Bovada, and BetRivers.
2. **Selenium web scraping** was used for DraftKings and BetMGM because these websites rely heavily on dynamically generated JavaScript content.

The system continuously collected data approximately every 90 seconds over multiple days. Upcoming games were prioritized because live betting odds change too rapidly and introduce additional instability into the arbitrage calculations. The scraping pipeline extracted:

- Team names
- Sportsbook name
- Decimal betting odds
- Event start time
- Sport category
- Timestamp of data collection

2.3 Challenges Encountered

Several challenges occurred during development:

- Different sportsbooks formatted team names differently.
- Some sportsbooks used accented characters such as “Montréal Canadiens.”
- Selenium scraping behaved differently in headless browser mode.
- Sportsbooks dynamically loaded odds data after page load.
- Odds APIs used different JSON structures than scraped websites.
- The biggest challenge involved maintaining Selenium scrapers over time.

To solve these problems, normalization functions were implemented to standardize team names and Unicode characters. Dynamic waits and scrolling logic were also added to Selenium scrapers to ensure all betting markets loaded correctly before extraction. However, no clear solution was found to maintain Selenium scrapers over time, since sports betting websites actively attempt to prevent automated scraping and frequently modify their HTML structure, CSS classes, and dynamically generated page elements. As a result, scraping code that functioned correctly one day could suddenly fail the next after a website update. This requires continuous debugging and adjustments to XPath selectors, CSS selectors, wait conditions, and page interaction logic in order to keep the data collection pipeline operational.

3 Integrated Data Model

The project integrated all sportsbook data into a unified tabular structure using Pandas DataFrames. Each sportsbook initially produced data in a different format, so cleaning functions standardized all records into the same schema. The primary integrated data model contained the following attributes:

Field	Description
sportsbook	Name of sportsbook
sport	NBA, MLB, or NHL
game_id	Standardized event identifier
team	Team name
odds_decimal	Decimal betting odds
commence_time	Game start time (UTC)
status	Upcoming or live
scrape_time	Timestamp of collection

The most important integration step was standardizing event identifiers across sportsbooks. Since sportsbooks used different naming conventions, normalized team names were combined into a canonical game identifier using sorted team names. For example:

lakers__warriors

This allowed odds from different sportsbooks to be merged together for arbitrage analysis.

3.1 Entity-Relationship Diagram

Figure 1 shows the informal entity-relationship diagram used to model the relationships between sportsbooks, sporting events, teams, and arbitrage opportunities.

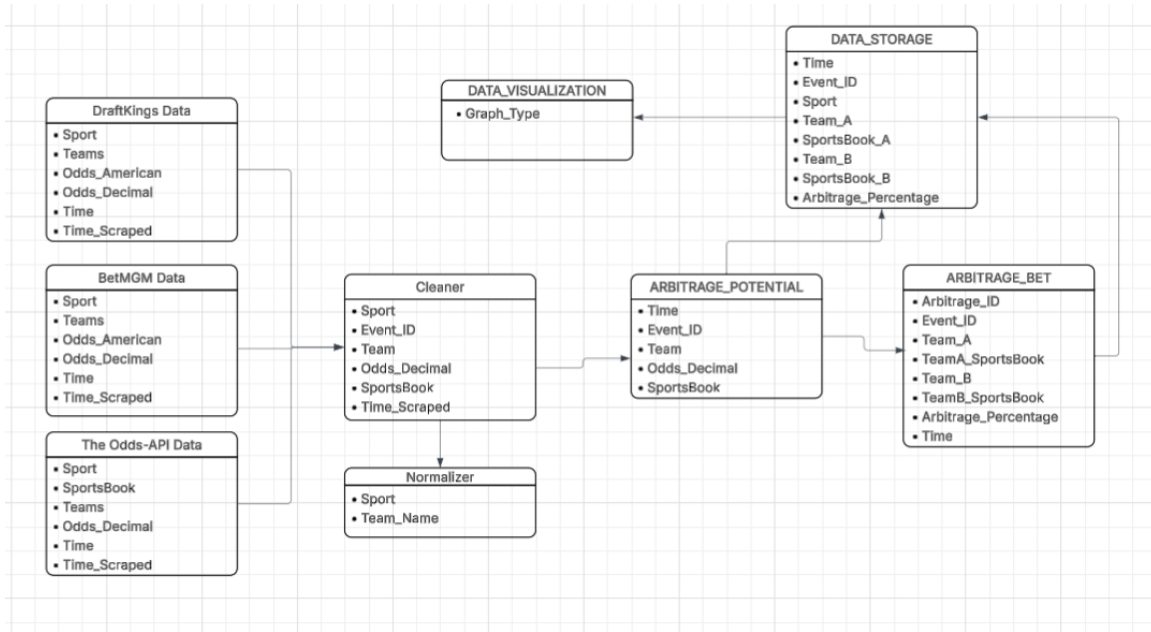


Figure 1: Informal Entity-Relationship Diagram

4 Analysis and Visualizations

After integration, the system analyzed the data to identify arbitrage opportunities. So for each sporting event, the system selected the best available odds for each possible outcome across all sportsbooks. Implied probabilities were then calculated using the formula:

$$\frac{1}{\text{Odds Team A}} + \frac{1}{\text{Odds Team B}}$$

If the sum of implied probabilities was less than 1, an arbitrage opportunity existed. Thus, the arbitrage profit margin was calculated as:

$$\text{Profit Margin} = \left(1 - \left(\frac{1}{\text{Odds Team A}} + \frac{1}{\text{Odds Team B}} \right) \right) \times 100$$

4.1 Frequency of Arbitrage Opportunities Over Time

Figure 1 shows how arbitrage opportunities fluctuated over time during data collection.

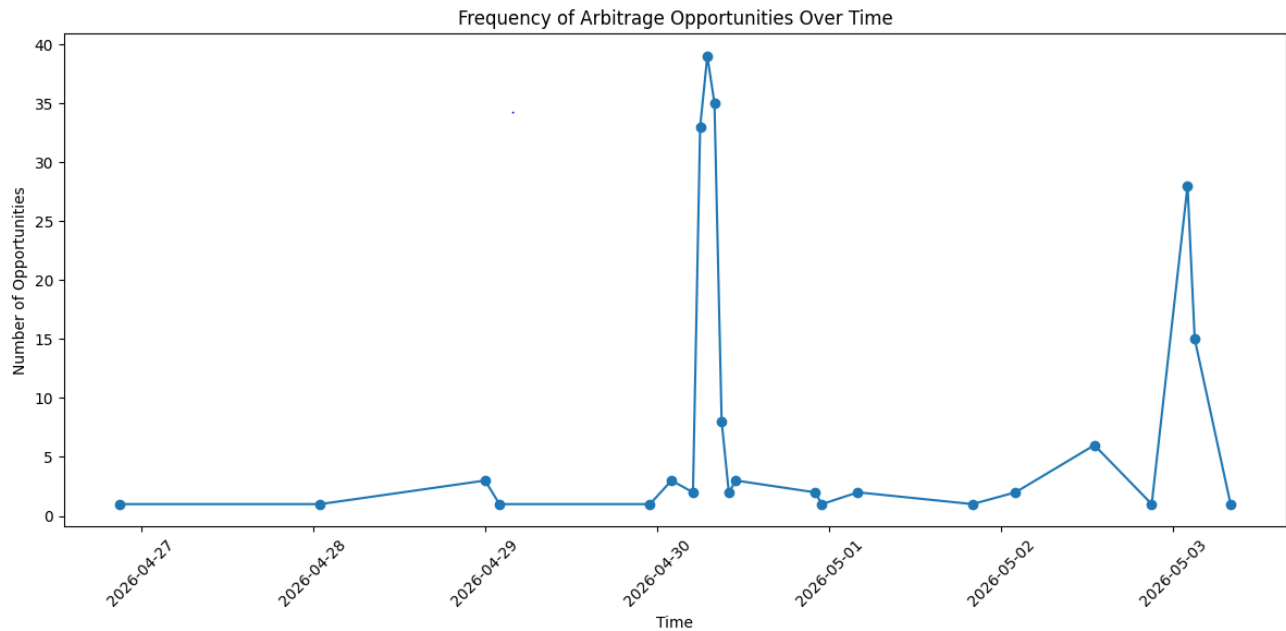


Figure 2: Frequency of Arbitrage Opportunities Over Time

The graph shows several large spikes where sportsbooks temporarily disagreed significantly on betting odds. These spikes suggest that arbitrage opportunities are highly time-dependent and occur in bursts rather than evenly over time.

4.2 Arbitrage Opportunities by Sport

Figure 2 compares the number of arbitrage opportunities discovered across different sports.

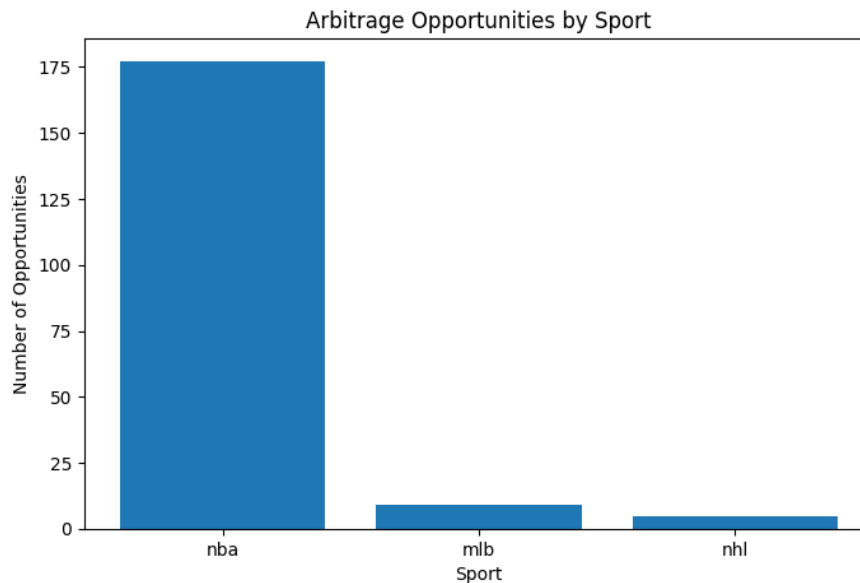


Figure 3: Arbitrage Opportunities by Sport

NBA games produced substantially more arbitrage opportunities than MLB or NHL. This may be because NBA games receive extremely high betting volume and sportsbooks adjust odds more aggressively and independently.

4.3 Distribution of Profit Margins

Figure 3 displays the distribution of arbitrage profit margins.

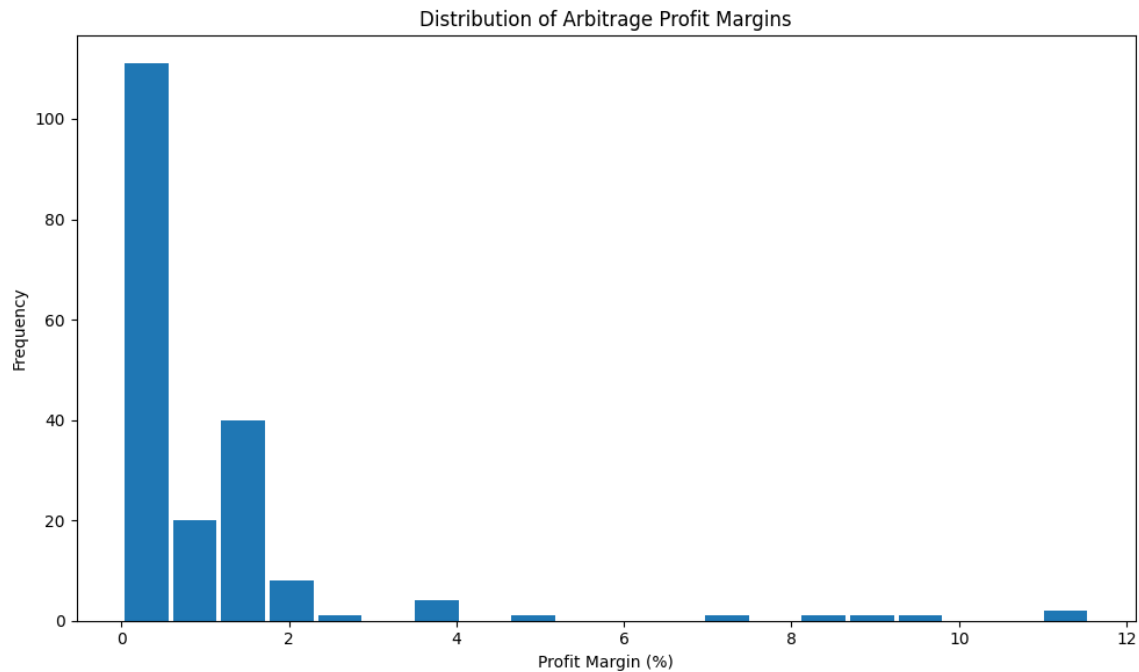


Figure 4: Distribution of Arbitrage Profit Margins

Most arbitrage opportunities had relatively small profit margins below 2%. However, several larger outliers were identified, including opportunities exceeding 10%. This demonstrates that while most arbitrage opportunities provide small returns, occasional pricing inefficiencies can create significantly larger profits.

4.4 Average Profit Margin by Sport

Figure 4 compares average profit margins by sport.

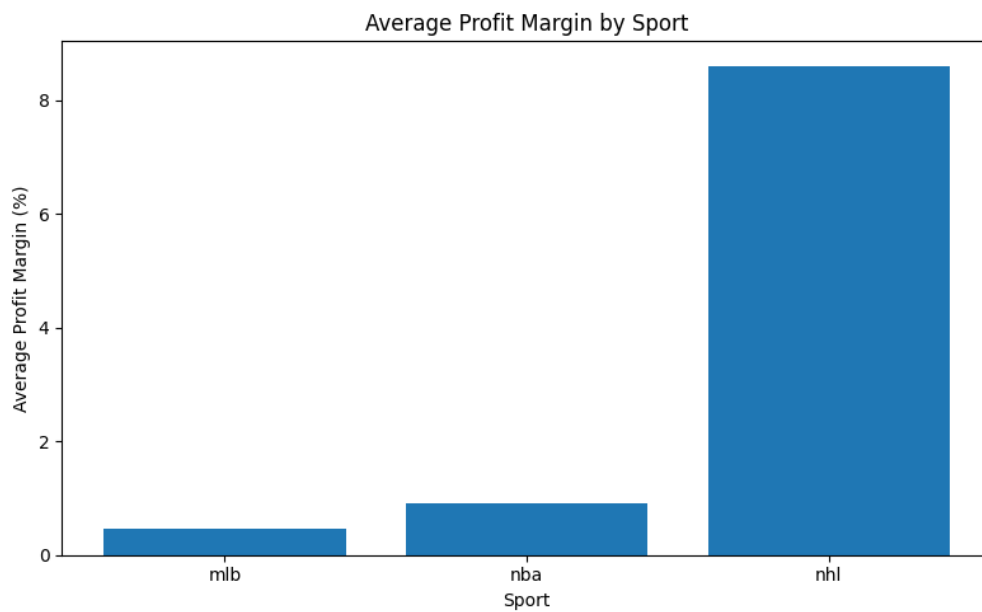


Figure 5: Average Profit Margin by Sport

Although NHL had fewer arbitrage opportunities overall, it produced the highest average profit margin. This suggests that less heavily bet sports may contain larger pricing inefficiencies.

4.5 Arbitrage Opportunities by Hour

Figure 5 shows how arbitrage opportunities varied throughout the day.

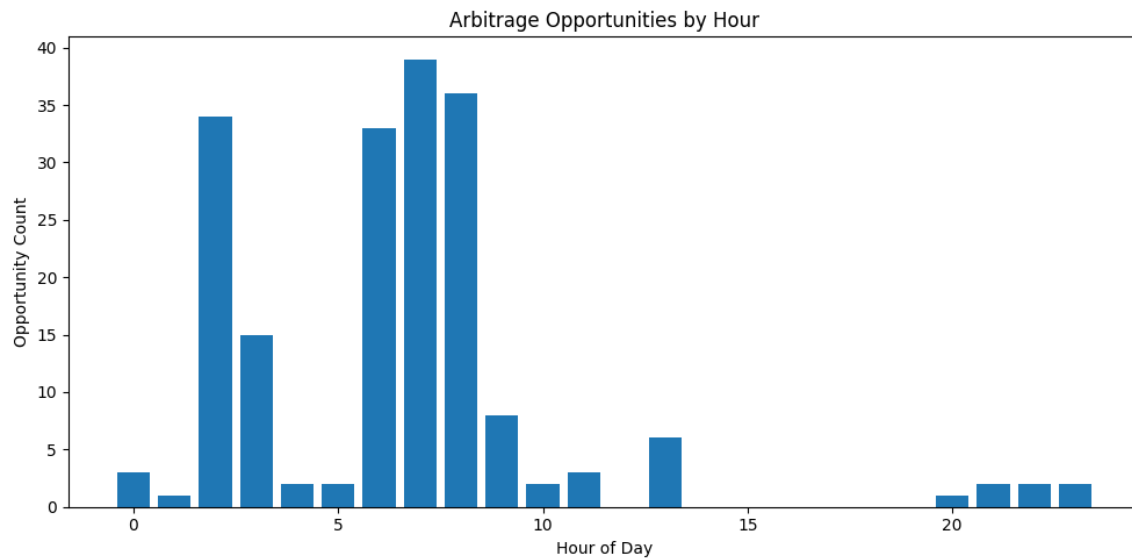


Figure 6: Arbitrage Opportunities by Hour

Most opportunities occurred during early morning hours and shortly before major sporting events began. This likely reflects periods where sportsbooks update odds asynchronously.

4.6 Top Sportsbook Pairings

Figure 6 identifies sportsbook combinations that produced the most arbitrage opportunities.

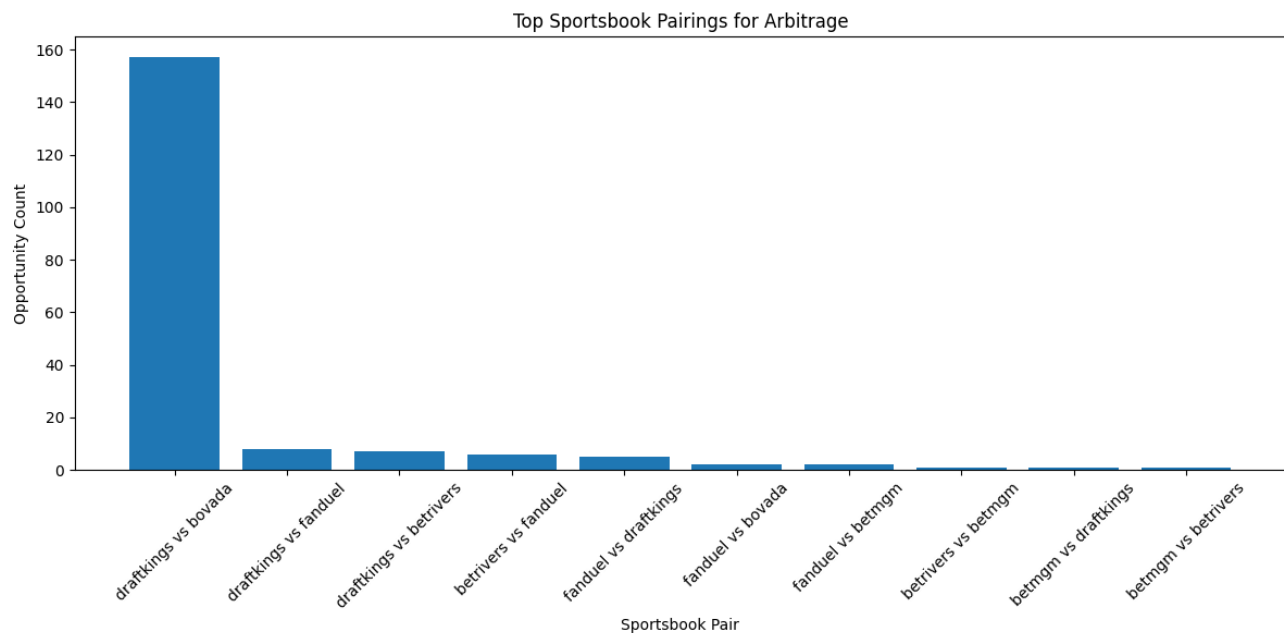


Figure 7: Top Sportsbook Pairings for Arbitrage

DraftKings and Bovada produced significantly more arbitrage opportunities than any other pairing. This suggests these sportsbooks price events differently more often than competitors.

4.7 Sportsbook Pair Heatmap

Figure 7 visualizes sportsbook pair relationships using a heatmap.

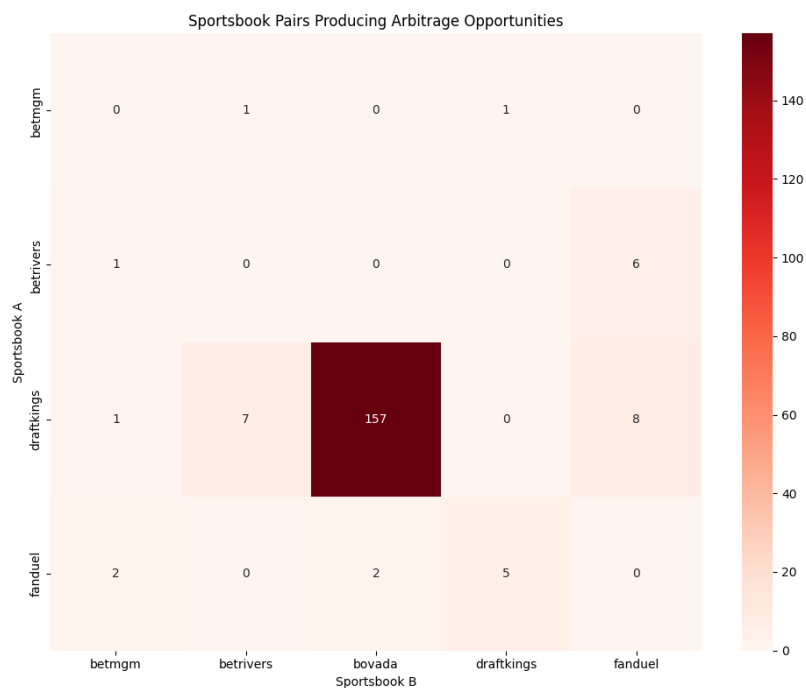


Figure 8: Sportsbook Pair Heatmap

The heatmap confirms that DraftKings and Bovada generated the highest concentration of arbitrage opportunities.

5 Conclusions

This project successfully demonstrated a complete end-to-end sports betting arbitrage detection pipeline. The system continuously collected sportsbook odds data, cleaned and normalized inconsistent records, integrated multiple sources into a unified dataset, and identified arbitrage opportunities automatically.

Several important findings emerged from the analysis:

- Arbitrage opportunities occur frequently but are highly time-sensitive.
- NBA generated the highest volume of opportunities.
- NHL opportunities produced the highest average profit margins.
- Most opportunities produced small profit percentages, but occasional large inefficiencies existed.
- DraftKings and Bovada produced the largest number of arbitrage pairings.

Overall, the project demonstrated how automated data collection and integration systems can identify meaningful inefficiencies in real-world markets.

6 Future Work

Given additional time, several improvements could be made to the project.

- Add support for additional sportsbooks.

- Expand to more sports and betting markets.
- Implement live betting arbitrage detection.
- Store historical data in a relational database.
- Build a real-time dashboard for monitoring opportunities.
- Add automated alert systems for profitable opportunities.
- Use machine learning models to predict when arbitrage opportunities are likely to occur.

Another future improvement would be optimizing Selenium scraping performance and implementing distributed data collection to increase collection frequency while minimizing API usage.